

Emergent maximum entropy production phenomena in viscoelastic medium

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Abstract

Active matter systems in crowded environments exhibit complex dynamics due to the interplay between energy injection and spatial caging. Here, we perform numerical simulations of a two-dimensional viscoelastic medium, modeled via overdamped Langevin dynamics with purely repulsive interactions, to investigate the system's response to an active periodic forcing. By varying the area fraction, we first characterize the passive transition from a fluid to a solid-like caged state. Subsequently, we show that when the external driving frequency exceeds the structural relaxation timescale of the dense medium, active particles spontaneously self-organize into clusters to overcome local elastic barriers. This collective yielding induces a rapid enhancement of the average dissipated power. Our results demonstrate that the driven bath dynamically restructures itself from a sterically frustrated state to maximize energy dissipation, providing evidence that emergent collective phenomena in this system are governed by the Maximum Entropy Production Principle (MEPP).

1. Introduction

In recent years, active matter systems have attracted interest from both the physics and biology communities due to their ability to extract energy from the environment and convert it into directed motion, driving themselves far from thermodynamic equilibrium [1]. The dynamics of passive Brownian particles is governed by thermal fluctuations arising from random collisions with surrounding fluid molecules; instead, active Brownian particles exhibit a complex interplay between thermal noise and active self-propulsion [2]. This phenomenon is ubiquitous in nature, from the macroscopic swarming of animal flocks [3] to the microscopic swimming of bacteria [1]. Furthermore, the development of artificial self-propelled nano- and micro-particles holds potential for technological applications, such as targeted drug delivery and bioremediation [2]. Because these microscopic entities operate in the low Reynolds number regime ($Re \ll 1$), inertial effects are negligible. Consequently, their dynamics are accurately described by overdamped Langevin equations, providing a robust theoretical and numerical framework to study how continuous injection of energy at the single-particle level leads to emergent collective behaviors on larger scales [4].

Although the behavior of isolated active particles in simple Newtonian fluids is well documented, real biological and technological environments are typically complex, crowded, and non-Newtonian. In such scenarios, the background medium cannot be modeled merely as a constant viscous drag. Instead, it exhibits viscoelastic properties characterized by transient structural memory and spatial caging effects [5]. To capture this physics, numerical experiments often employ dense suspensions of particles interacting via purely repulsive steric forces, such as the Weeks-Chandler-Andersen (WCA) potential. In these colloidal baths, an active "probe" particle subjected to a periodic external force (e.g., $F_{ext} \propto \sin(\omega t)$)

must overcome the local elastic cage formed by its neighbors. Consequently, the macroscopic structural response of the system is deeply governed by the interplay between the density of the bath and the forcing parameters (amplitude and frequency).

Within the framework of overdamped Langevin dynamics, the mechanical work performed by non-conservative external forces is entirely dissipated as heat into the surrounding thermal bath [6]. Consequently, measuring the average dissipated power yields a direct estimation of the entropy production rate in the non-equilibrium steady state. In this work, we present numerical simulations of a two-dimensional WCA fluid subjected to an active periodic forcing. By systematically varying the amplitude of the drive, we aim to explore the resulting dynamical regimes and investigate whether the emergence of collective behavior is guided by the MEPP [7].

2. Theoretical framework and simulation details

2.1. System

We consider a two-dimensional system of interacting particles confined in a square box of side L . The number density of the system is defined as $\rho = N/L^2$. In our numerical experiments, the crowding of the environment is parameterized by the dimensionless area fraction ϕ , which is geometrically related to the number density by the relation $\phi = \rho\pi\sigma^2/4$, where σ represents the characteristic diameter of the particle. Since the total number of particles N is kept constant, exploring different area fractions ϕ intrinsically requires scaling the dimension of the box L . To minimize finite-size effects and accurately simulate a bulk system, standard periodic boundary conditions (PBC) are applied along both Cartesian axes [8]. To ensure statistical robustness, all quantitative data and ensemble averages were calculated using a reference configuration of $N = 529$ particles. This system size rep-

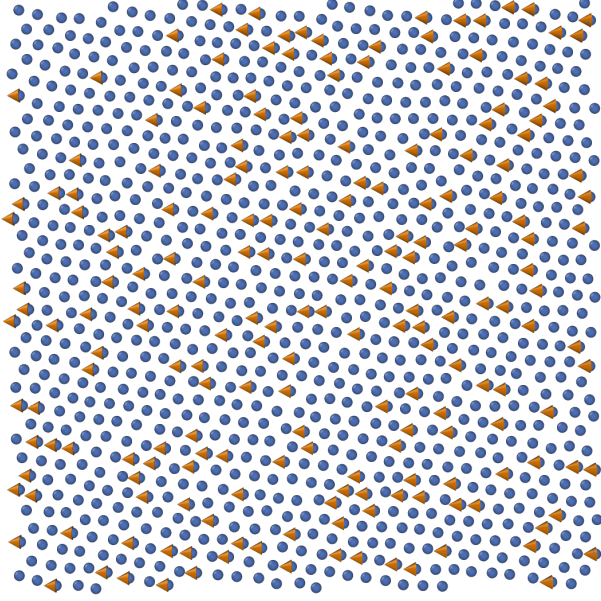


Figure 1: Initial configuration of the particles in the active regime (after thermalization). Blue denotes passive particles and orange active ones. For visual purposes a different number of particles $N = 1024$ was chosen.

resents a balanced compromise to optimize the trade-off between computational feasibility and the requirement for a representative thermodynamic environment. At the highest density investigated, the linear dimension of the simulation box reaches its minimum value ($L \approx 23\sigma$), but it remains significantly larger than the typical spatial correlation length λ of the system. This ensures that even in the most tightly packed regime, intensive observables are representative of the thermodynamic limit. In contrast, snapshots for visual analysis were generated using a larger system of $N = 1024$ particles (Figure 1). This choice was made to provide a clearer morphological representation of collective phenomena by minimizing visual artifacts arising from periodic boundary conditions and allowing for a more natural development of spatial patterns. We verified that qualitative behavior and structural properties, such as the radial distribution function $g(r)$, remain consistent between the two sizes of the system.

2.2. Model

2.2.1. Overdamped Langevin dynamics

The time evolution of each particle i at position $\mathbf{r}_i$ is governed by the overdamped Langevin stochastic differential equation (SDE). In our model, we consider a dense colloidal suspension where inertial effects and hydrodynamic interactions are completely neglected. Setting the mobility $\mu = 1.0$ and working in reduced units where the Boltzmann constant is $k_B = 1$, the viscous friction coefficient becomes $\gamma = 1/\mu = 1$, and the translational diffusion coefficient is given by Einstein's relation $D = \mu k_B T = T$. For the purely passive system, the

resulting SDE is

$$\dot{\mathbf{r}}_i = \mu \mathbf{F}_i^{WCA} + \boldsymbol{\eta}_i(t). \quad (1)$$

The term $\boldsymbol{\eta}_i(t)$ represents a Gaussian white noise that models stochastic thermal motion. It satisfies the fluctuation-dissipation theorem, characterized by a zero mean and a variance proportional to the temperature

$$\langle \boldsymbol{\eta}_i(t) \rangle = 0 \quad \text{and} \quad \langle \eta_{i,\alpha}(t) \eta_{j,\beta}(t') \rangle = 2D \delta_{ij} \delta_{\alpha\beta} \delta(t - t')$$

where α and β denote Cartesian coordinates.

2.2.2. Steric interactions

To model excluded volume effects without introducing attractive cohesive forces, particles interact pairwise via the WCA potential [9]. This is defined as a standard Lennard-Jones potential truncated and shifted at its absolute minimum

$$U_{WCA}(r) = \begin{cases} 4\epsilon[(\sigma/r)^{12} - (\sigma/r)^6] + \epsilon, & r \leq 2^{1/6}\sigma \\ 0, & r > 2^{1/6}\sigma \end{cases} \quad (2)$$

where $r = |\mathbf{r}_i - \mathbf{r}_j|$ is the distance between the particles. Without loss of generality, we set the energy scale $\epsilon = 1$ and the radius of the particles $\sigma = 1$. The interaction cutoff point is strictly set at $r_c = 2^{1/6}\sigma \approx 1.12246$. The use of the WCA potential is justified by the fact that the local structural ordering of dense liquid and viscoelastic environments is predominantly dictated by short-range repulsive steric forces rather than attractive tails [9]. The total repulsive force acting on the particle i is analytically evaluated as the negative gradient of the potential

$$\mathbf{F}_i^{WCA} = - \sum_{j \neq i} \nabla_{\mathbf{r}_i} U_{WCA}(|\mathbf{r}_i - \mathbf{r}_j|).$$

2.2.3. Active particles

To drive the system out of equilibrium, 20% of the particles are designated as active. Unlike purely passive Brownian particles, these active probes are subjected to an external, time-dependent periodic force directed along the x -axis. The active forcing term is mathematically defined as

$$\mathbf{F}_i^{ext}(t) = A \sin(\omega t) \hat{x} \quad \text{for } i \in \text{active set.}$$

Consequently, the continuous overdamped Langevin equation for this subset of particles is modified to include the external drive

$$\dot{\mathbf{r}}_i = \mu (\mathbf{F}_i^{WCA} + \mathbf{F}_i^{ext}(t)) + \boldsymbol{\eta}_i(t). \quad (3)$$

From a physical standpoint, the amplitude A dictates the peak mechanical intensity of the active particle, representing the maximum energy injected into the system to overcome the local viscoelastic cage. The angular frequency ω establishes the timescale over which the particle reverses its propulsion direction.

2.3. Numerical details

2.3.1. Integration scheme and boundary conditions

The continuous overdamped Langevin equations are numerically integrated using the explicit Euler-Maruyama method [10]. This first-order stochastic scheme is the standard choice for Brownian dynamics, as higher-order integrators do not provide significant advantages due to the low temporal regularity of the Wiener process. By integrating deterministic forces and stochastic white noise over a discrete time-step Δt , the updated position of a passive particle at time $t + \Delta t$ is computed as

$$\mathbf{r}_i(t + \Delta t) = \mathbf{r}_i(t) + \mu \mathbf{F}_i^{WCA}(t) \Delta t + \sqrt{2D\Delta t} \boldsymbol{\xi}_i. \quad (4)$$

For active particles, the position is updated as

$$\mathbf{r}_i(t + \Delta t) = \mathbf{r}_i(t) + \mu [\mathbf{F}_i^{WCA}(t) + \mathbf{F}_i^{ext}(t)] \Delta t + \sqrt{2D\Delta t} \boldsymbol{\xi}_i, \quad (5)$$

where $\boldsymbol{\xi}_i$ is a vector of independent random variables drawn from a standard normal distribution, i.e., $\xi_{i,\alpha} \sim \mathcal{N}(0, 1)$.

A critical numerical parameter in this scheme is the integration time-step, Δt . Because the WCA potential features a highly repulsive r^{-12} core, particles experiencing close-contact collisions are subjected to steeply diverging forces. To ensure numerical stability and prevent non-physical coordinate overlaps, we employ a conservative time-step of $\Delta t = 10^{-4}$ in reduced units.

To simulate a bulk system and minimize finite-size effects, the integration is performed within a square box equipped with Periodic Boundary Conditions (PBC). The Minimum Image Convention is rigorously applied when computing both interparticle distances r_{ij} and the corresponding pairwise forces. To reduce the computational complexity from $\mathcal{O}(N^2)$ to $\mathcal{O}(N)$, the force evaluation relies on a Cell Lists algorithm, with a grid size strictly larger than the interaction cutoff r_c [8].

2.3.2. Equilibration

The simulation protocol begins with the generation of the initial spatial coordinates. To avoid infinite repulsive forces at $t = 0$, particles are not placed randomly; instead, they are initialized on a perfect square lattice [8]. The spacing of the lattice and the size of the box L are analytically determined by a custom Python script, given the total number of particles N and the target packing fraction $\phi = N\pi(\sigma/2)^2/L^2$. Because the perfectly ordered lattice is an artificial state that does not reflect the equilibrium structure of a dense fluid, the system must be dynamically relaxed before any data acquisition takes place.

To monitor the thermodynamic state and ensure proper equilibration, the potential energy per particle, U/N , is evaluated throughout the simulation trajectory. This macroscopic observable is calculated by summing the pairwise WCA interactions over all particles separated by a distance smaller than the cutoff radius r_c . Monitoring the temporal evolution of U/N is crucial during

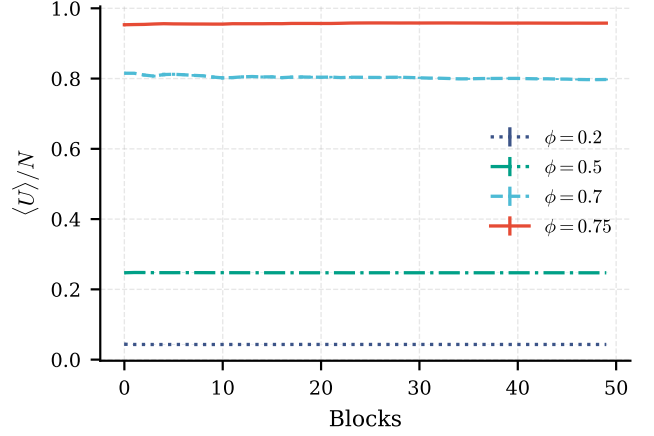


Figure 2: Average potential energy (per particle) as a function of the cumulative block index for different area fractions ϕ , in the passive regime. Computed using block averaging (Section 2.4.1).

the initial thermalization phase: it provides a quantitative metric to confirm the melting of the artificial initial configuration. It ensures that the system has reached a steady state. Only after this steady state has been unequivocally attained are dynamical observables extracted to characterize the transport and rheological properties of the medium.

From a physical perspective, U/N serves as an excellent probe for equilibration because it directly reflects the local coordination shell of the particles. During the early stages of the simulation, the initial crystalline constraint forces the particles into suboptimal relative distances, generating high-energy steric overlaps. As the lattice melts due to thermal fluctuations, the particles rearrange to find local potential minima, causing a rapid variation in the internal energy. Once U/N stops exhibiting drifts and merely fluctuates around a well-defined stationary mean, it indicates that the structural memory of the initial state has been completely erased (Figure 2). In our standard protocol, a thermalization phase of 10^5 integration steps proves sufficient to reach this stationary state.

2.3.3. Simulation parameters and protocol

To summarize the standard protocol, all simulations are performed using $N = 529$ particles in a two-dimensional domain with periodic boundary conditions (PBC). We operate in standard reduced units, defining the WCA energy scale $\epsilon = 1$, the diameter of the particle $\sigma = 1$, and the thermal energy $k_B T = 1$. To ensure that the results are not influenced by the initial artificial configuration, the system is first thermalized for 10^5 steps, allowing it to reach a stable thermodynamic steady-state. Following this equilibration and a subsequent characterization of the passive dynamics, active driving is introduced by assigning a periodic forcing $F_{ext}(t) = A \sin(\omega t)$ to 20% of the total particles. To avoid transients or contamination from the initial setup, the active simulation is initialized directly from the final configuration of an already equilibrated passive system. This protocol ensures

that the system has properly relaxed from the initial spatial configuration before the external driving force is applied. Finally, the data production phase spans $5 \cdot 10^6$ total steps, ensuring robust statistics for the calculation of thermodynamic averages and mean squared displacement.

2.4. Analysis

2.4.1. Data blocking

Even in the steady state, extracting reliable statistical uncertainties from a Markov chain is non-trivial due to temporal autocorrelations. Two measurements taken at contiguous time-steps are highly correlated and do not constitute independent statistical samples. To rigorously estimate the standard errors of the macroscopic observables, we employ the standard block averaging technique [11].

The production trajectory is partitioned *a priori* into N_{blocks} independent blocks of equal length, each containing N_{steps} integration steps. The length of the block ($T_{block} = N_{steps}\Delta t$) must be chosen to be significantly larger than the longest structural relaxation time of the system, ensuring that the local block averages behave as statistically uncorrelated and independent variables [12]. To properly quantify this relaxation time, we computed the system's autocorrelation function over a trajectory of $2 \cdot 10^5$ steps (after thermalization), identifying the decorrelation time as the first instance the autocorrelation drops below a threshold of 5×10^{-4} . This calibration procedure was performed exclusively at the maximum area fraction considered ($\phi = 0.75$). Since structural relaxation slows down and autocorrelation times become naturally longer at higher densities, a block size appropriately chosen for $\phi = 0.75$ intrinsically guarantees statistical independence for all lower density regimes. In our standard protocol, we operate with $N_{blocks} = 50$ and $N_{steps} = 10^5$. For each block k , a local average of the observable is computed on-the-fly. The final estimate is the global mean of the block averages, and the true statistical error is evaluated as the standard error of the mean across the N_{blocks} independent data points.

This block partitioning imposes a strict mathematical constraint on the external driving frequency, ω . To ensure a consistent time-average of periodic observables (such as the dissipated power), the total duration of a single block, $T_{block} = N_{steps}\Delta t$, must contain an exact integer number of forcing periods, $T_{ext} = 2\pi/\omega$. Therefore, the chosen frequencies must satisfy the resonance condition

$$\omega_m = m \frac{2\pi}{N_{steps}\Delta t}, \quad \text{with } m \in N^+. \quad (6)$$

If this condition is violated, the integration over the block captures fractional parts of a phase, introducing an artificial, non-zero bias in the block average of the dissipated energy.

2.4.2. Radial distribution function

To characterize the spatial organization and the local structural ordering of the dense colloidal bath, we compute the radial distribution function $g(r)$. This static observable quantifies the probability of finding a particle at a radial distance r from a reference particle, normalized by the probability expected for a completely uncorrelated ideal gas at the same uniform density ρ [9].

For a two-dimensional system of N particles in an area $V = L^2$, the mathematical definition of $g(r)$ is given by

$$g(r) = \frac{1}{\rho^2 V} \left\langle \sum_{i=1}^N \sum_{j \neq i}^N \delta(\mathbf{r} - \mathbf{r}_i + \mathbf{r}_j) \right\rangle. \quad (7)$$

Numerically, this continuous probability density is evaluated by partitioning the radial distance into discrete concentric annuli of thickness Δr . At each sampling step, the number of particle pairs separated by a distance within the interval $[r, r + \Delta r]$ is counted. Following the application of periodic boundary conditions, the radial distance is restricted to the maximum reliable scale, $r < L/2$, to prevent artifactual self-correlations. The calculation is performed following the standard histogram-based approach described in [8]. The analysis of the peaks in the $g(r)$ allows us to unambiguously distinguish between the completely disordered state characteristic of a dilute fluid and the short-range local ordering typical of a dense viscoelastic medium or a glass-forming liquid (Figure 3).

2.4.3. Mean squared displacement

The Mean Squared Displacement (MSD; Figure 4) constitutes the fundamental dynamical observable to classify the transport regime of the medium (e.g., diffusive, subdiffusive). To prevent artificial truncations introduced by the periodic boundary conditions, the absolute particle paths are reconstructed by tracking the *unwrapped* coordinates, unlike *wrapped* coordinates, which are artificially mapped back into the primary simulation box to correctly evaluate interparticle forces. Unwrapped coordinates represent the continuous and unconstrained spatial evolution of a particle in an infinite domain. Tracking this unconstrained motion is strictly necessary to correctly quantify long-range diffusion [8].

Extracting dynamical properties from a finite number of simulated particles requires maximizing statistical accuracy. This is achieved by computing the Time-Averaged Mean Squared Displacement (TAMSD). For a single particle trajectory $\mathbf{r}_i(t)$ recorded over a total time t_{max} , TAMSD as a function of the lag-time τ is defined as

$$\overline{\delta_i^2(\tau)} = \frac{1}{t_{max} - \tau} \int_0^{t_{max}-\tau} |\mathbf{r}_i(t + \tau) - \mathbf{r}_i(t)|^2 dt. \quad (8)$$

Numerically, since particle trajectories are recorded as a finite sequence of discrete simulation frames, the continuous time integral is evaluated as an arithmetic mean over all valid temporal origins for a given lag-time τ . Physically, this operation involves sliding a time window of length τ across the entire observation period t_{max} .

Furthermore, to ensure robust statistical sampling and mitigate the characteristic noise divergence at large lag-times, the TAMSD computation is strictly evaluated up to a maximum lag-time corresponding to 80% of the total trajectory length. This sliding-window approach significantly increases the statistical weight of the measurement without requiring computationally expensive independent simulation runs.

Considering the system as ergodic, the ensemble average can be safely combined with the time average [12]. Thus, the overall MSD of the system is evaluated by averaging the TAMSD over all N particles

$$\langle \Delta r^2(\tau) \rangle = \frac{1}{N} \sum_{i=1}^N \overline{\delta_i^2(\tau)}. \quad (9)$$

This dual averaging smooths thermal fluctuations and provides a highly reliable statistical estimator.

2.4.4. Storage and Loss moduli

The macroscopic linear viscoelasticity of the passive bath is quantified through frequency-dependent complex shear modulus $G^*(\omega) = G'(\omega) + iG''(\omega)$. The real part, $G'(\omega)$, is the storage modulus and represents the elastic solid-like response; the imaginary part, $G''(\omega)$, is the loss modulus and accounts for the viscous energy dissipation. These macroscopic moduli are related to microscopic particle dynamics via the Generalized Stokes-Einstein Relation (GSER), which analytically involves the unilateral Laplace transform of the MSD.

Directly computing the Laplace transform of discrete, finite numerical data is mathematically ill-posed and highly susceptible to truncation errors. To circumvent this instability in our analysis, we utilize a robust algebraic approximation based on the local logarithmic derivative of the MSD [13]. We evaluate the local power-law exponent as

$$\alpha(\omega) = \left. \frac{d \ln \langle \Delta r^2(t) \rangle}{d \ln t} \right|_{t=1/\omega}. \quad (10)$$

By locally approximating the MSD as a power law, $\langle \Delta r^2(t) \rangle \propto t^\alpha$, its Laplace transform can be evaluated analytically. The integral of a power law in the Laplace domain naturally yields the Gamma function, $\Gamma(1 + \alpha)$, extending the factorial to real numbers. Evaluating the GSER with this substitution yields the magnitude of the complex modulus

$$|G^*(\omega)| \approx \frac{k_B T}{\pi a \langle \Delta r^2(1/\omega) \rangle \Gamma[1 + \alpha(\omega)]}, \quad (11)$$

where a represents the characteristic length scale of the probe particle. Applying Euler's formula to separate the real and imaginary components based on the phase angle dictated by $\alpha(\omega)$, the storage and loss moduli are respectively extracted as

$$G'(\omega) = |G^*(\omega)| \cos\left(\frac{\pi}{2} \alpha(\omega)\right), \quad (12)$$

$$G''(\omega) = |G^*(\omega)| \sin\left(\frac{\pi}{2} \alpha(\omega)\right). \quad (13)$$

This algebraic method completely bypasses numerical integration, mapping the instantaneous slope of the statistically averaged MSD in logarithmic scale directly to the viscoelastic phase angle.

2.4.5. Entropy production and dissipated power

A hallmark of active matter systems is the continuous injection of energy at the single-particle level, which breaks detailed balance and drives the system into a non-equilibrium steady state (NESS). Within the framework of stochastic thermodynamics [6], the energetics of such systems can be evaluated directly along fluctuating particle trajectories.

For the active subset of particles in our model, the external periodic driving force, $\mathbf{F}_i^{ext}(t)$, performs mechanical work. The instantaneous rate of this work, or injected power, for a single active particle i moving with velocity $\mathbf{v}_i(t)$ is

$$\dot{W}_i(t) = \mathbf{F}_i^{ext}(t) \cdot \mathbf{v}_i(t). \quad (14)$$

In the overdamped limit, inertial effects are strictly negligible, implying no mechanical energy is stored as kinetic energy. According to the energetics of fluctuating trajectories [14], the instantaneous work dW_i performed by the active forcing is partitioned into the heat dQ_i dissipated into the thermal bath and the infinitesimal change in the internal potential energy dU_i . This yields the stochastic first law at the single-particle level: $dW_i = dQ_i + dU_i$. Because the total internal potential energy (governed by the WCA interactions) is bounded, its time derivative strictly vanishes upon time-averaging in the NESS ($\langle \dot{U}_i \rangle = 0$). Consequently, the mechanical power injected by the external drive is, on average, entirely dissipated as heat into the environment, leading to the macroscopic balance $\langle \dot{Q} \rangle = \langle \dot{W} \rangle$.

Assuming the implicit solvent acts as an ideal thermal bath at a constant temperature T , this heat dissipation generates a corresponding entropy flux. The mean entropy production rate of the system, $\langle \dot{S} \rangle$, maps directly to the average dissipated power

$$\langle \dot{S} \rangle = \frac{\langle \dot{Q} \rangle}{T} = \frac{1}{TN} \sum_{i \in \text{active}} \langle \mathbf{F}_i^{ext}(t) \cdot \mathbf{v}_i(t) \rangle. \quad (15)$$

Evaluating this quantity allows us to investigate the thermodynamic cost of structural rearrangements within the viscoelastic bath and to test whether emergent collective phenomena are governed by a principle of maximum entropy production [7].

3. Results

3.1. Characterization of the passive viscoelastic medium

Before investigating the non-equilibrium dynamics induced by the active forcing, it is necessary to establish a baseline reference of the purely passive system for subsequent analysis. After the system is thermalized for 10^5 steps, to ensure an ergodic initial condition, the primary

objective is to quantify how the dense packing and the WCA repulsive potential dictate the structural and mechanical properties of the medium. We first investigate the spatial organization and dynamical behavior of the system to characterize the viscoelastic moduli and determine the structural relaxation times of the background phase.

3.1.1. Structural analysis

The spatial ordering of the passive medium is characterized by the radial density function $g(r)$, which provides a statistical measure of the local density variations as a function of the distance r from a reference particle (Section 2.4.2).

We investigated the structural behavior of the system across different area fractions ϕ , revealing a transition from a liquid-like state to a highly dense environment. For all the investigated densities, the radial distribution function $g(r)$ exhibits an excluded-volume region at short distances where $g(r) \approx 0$, a direct consequence of the steeply repulsive core of the WCA potential. A pronounced first peak emerges at $r \approx 2^{1/6}\sigma$, identifying the first coordination shell of closely packed neighboring particles. Furthermore, a second peak is observed at $r \approx 2\sigma$, indicating the formation of a second coordination shell, followed by the typical damped oscillations characteristic of short-range structural order.

As can be seen in Figure 3, at $\phi = 0.2$ the $g(r)$ remains nearly featureless. Beyond the excluded volume region ($r < 2^{1/6}\sigma$) the function rapidly plateaus to unity. The absence of secondary peaks indicates that the inter-particle distances are large enough that the particles do not form stable coordination shells. This confirms that at low density, the system lacks long-range structural correlations and behaves as a simple Newtonian fluid. As the area fraction increases to $\phi = 0.7$, we observe a sharp first peak and the formation of secondary peaks that denote the aggregation in coordinated shells of the particles.

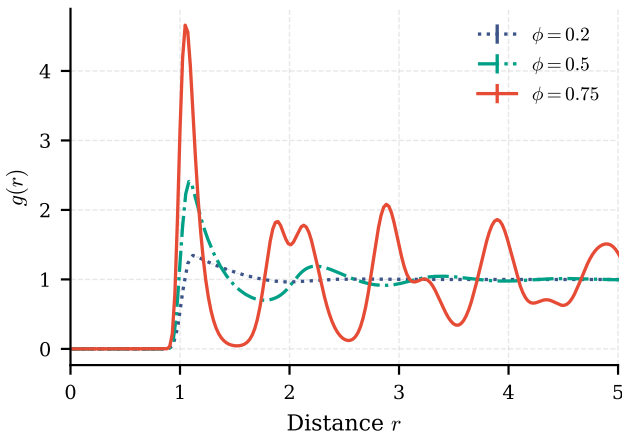


Figure 3: Radial distribution function computed for different area fractions in the passive regime.

At the highest area fraction $\phi = 0.75$ the $g(r)$ exhibits high-amplitude, narrow peaks. The persistence of these

peaks shows how each particle is locked in a structural cage by its neighbors. The primary peak is significantly more intense, reflecting the high probability of finding neighbors at the steric limit. The second peak is split in two, caused by particles at distance 2σ from the reference aligning either in a straight line or in a triangular/hexagonal pocket. Even though the medium is not a crystal (it lacks long-range order), this denotes a very specific local geometry.

3.1.2. Dynamical regime

To investigate the dynamical properties of the passive medium and the underlying relaxation process, we study the MSD (Section 2.4.3), computed using unwrapped trajectories to account for long-range diffusion.

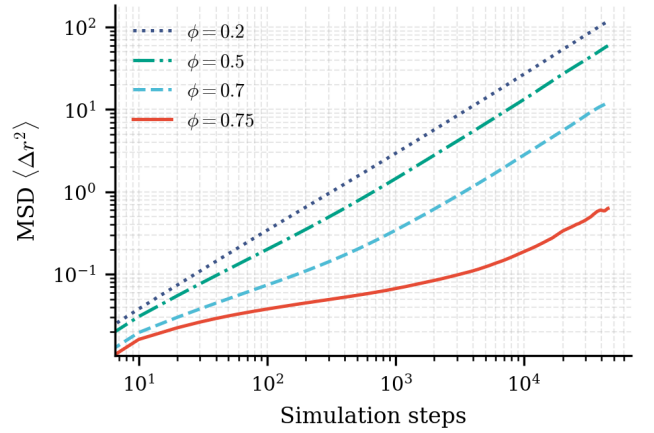


Figure 4: Time-Averaged Mean Squared Displacement as function of simulation steps, for different area fractions in the passive regime.

The dynamical behavior of the system is highly dependent on the area fraction; this justifies the choice of four area fractions ϕ , which represent different domains of the system.

For low densities, the system is in a dilute regime, where particles are spatially separated and collisions are rare; here, the MSD takes an almost-linear growth across all timescales, $\langle \delta r^2(\tau) \rangle \sim \tau$. As can be seen in Figure 4, the log-log curve of the $\phi = 0.2$ density is almost linear, with the local power-law exponent α estimated to be in the interval from 0.95 to 1, indicating that the medium behaves as a simple Newtonian fluid.

As the density of the system increases, the influence of the WCA potential becomes prominent and, while the short-term behavior remains diffusive, at intermediate scales we observe a sub-diffusive regime where α drops below 1. This occurrence shows that the particles start to feel the structural constraints of their neighbors, even though they have enough energy to escape these transient cages quickly and re-establish a diffusive regime, as can be seen in Figure 4 from the $\phi = 0.7$ curve.

Finally, for high densities, the system shows a caged behavior, where the MSD shows a long sub-diffusive plateau which spans several timescales. In this regime,

the particles are effectively trapped by their neighbors and the MSD remains approximately constant, with variations reflecting the characteristic length scale of the cages. Eventually, after a delay much longer than at lower densities, the system restores the diffusive regime when the particles escape their cages. This behavior can be correctly observed in the $\phi = 0.75$ curve of Figure 4.

3.1.3. Microrheology and the viscoelastic transition

To quantify the macroscopic linear viscoelasticity of the passive bath, we extracted the frequency-dependent storage and loss moduli, $G'(\omega)$ and $G''(\omega)$, using the Mason-Weitz approximation (Section 2.4.4). For clarity of presentation, Figure 5 displays the results for only two representative area fractions, $\phi = 0.5$ and $\phi = 0.75$. These specific densities were selected because they optimally illustrate the structural transition in the medium's response to mechanical stress without cluttering the plot. Specifically, $\phi = 0.5$ exemplifies the typical fluid-like regime—where the moduli do not intersect and the viscous response dominates—whereas $\phi = 0.75$ captures the emergence of solid-like viscoelastic features characteristic of a highly crowded environment.

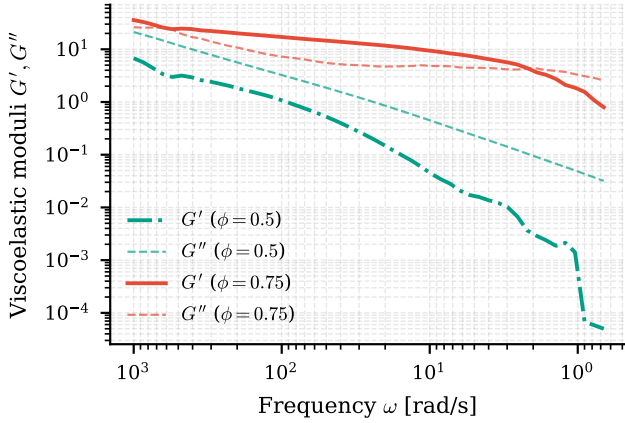


Figure 5: Storage (G') and Loss (G'') moduli computed for four different area fractions in the passive regime, using the Mason-Weitz approximation [13].

In particular, at low density the loss modulus G'' (dashed line) significantly dominates the storage modulus G' (solid line), confirming that in the dilute regime the system behaves approximately as a fluid across the entire frequency spectrum considered and any mechanical work injected is dissipated as heat. At high density instead, we observe a transition towards a solid-like regime: at low frequencies the system remains dominated by the viscous behavior, but at higher frequencies we observe a crossover to a regime where $G' > G''$ at $\omega_c \approx 2$. This dominance is a signature of caging, since at these timescales the particles are elastically trapped by their neighbors and cannot relax structurally, causing the medium to store energy rather than dissipate it.

3.2. Active particles behavior in viscoelastic matrix

Once the passive bath is fully characterized, we turn to investigating the system's response to the continuous

injection of mechanical work in the medium by introducing the active driving component: we assign a periodic external force $F_{ext}(t) = A \sin(\omega t)$ to 20% of the particles (for details see Section 2.2.3). The amplitude A of the forcing dictates the intensity of energy injected into the medium.

3.2.1. Collective organization

One of the core findings of our simulations is the emergence of collective organization among active particles as the driving amplitude A increases, specifically for forcing frequencies above the crossover ω_c . In the following, we consider $\omega \approx 10$ and an area fraction $\phi = 0.75$.

To analyze the structural response of the system, we introduce the Péclet number Pe , which quantifies the ratio between active forcing transport and stochastic thermal diffusion: $Pe = v_c \sigma / D$. It represents the competition between the external work injected and the caging of the passive medium: a higher value indicates that active particles possess sufficient energy to overcome the local cages and aggregate. The characteristic speed of the forcing can be derived directly from the Langevin equations (3) by ignoring thermal noise and the interaction term, so that $v_c = A/\gamma$; on the other hand, D is the diffusion coefficient, given by Einstein's relation $D = \mu k_B T$. Therefore, considering natural units, we obtain that the dimensionless Péclet number equals the forcing amplitude: $Pe = A$. As the driving amplitude

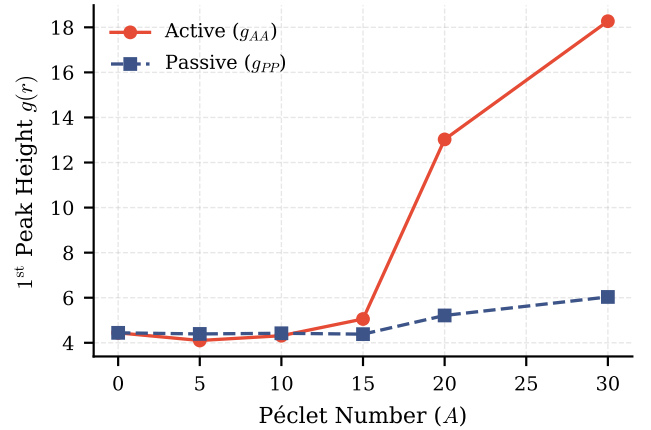


Figure 6: Comparison of the radial density function's first peak height against the Péclet Number, for active and passive particles, in the active regime.

A increases, we compare the height of the first peak of the radial distribution function for active-active pairs (g_{AA}) and passive-passive pairs (g_{PP}), which serves as an indicator of the system's caging. In Figure 6, we observe that the structural configuration of the passive medium (represented by the g_{PP} curve) remains approximately unchanged. On the other hand, for low Péclet numbers, the active forcing is unable to break the cages of the passive medium; however, above a certain threshold, the height of the active curve g_{AA} increases dramatically. This marks a dynamical transition to a state where active particles are able to break the cages and form clusters.

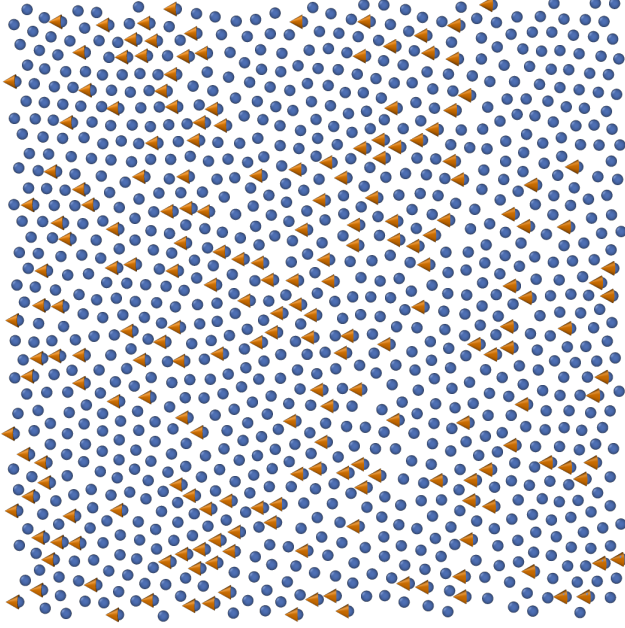


Figure 7: Visualization of liquid-like regime. Active particles (in orange) have a sinusoidal forcing of frequency $\omega \approx 0.63$ under the crossover ω_c . For visual purposes a different number of particles $N = 1024$ was chosen.

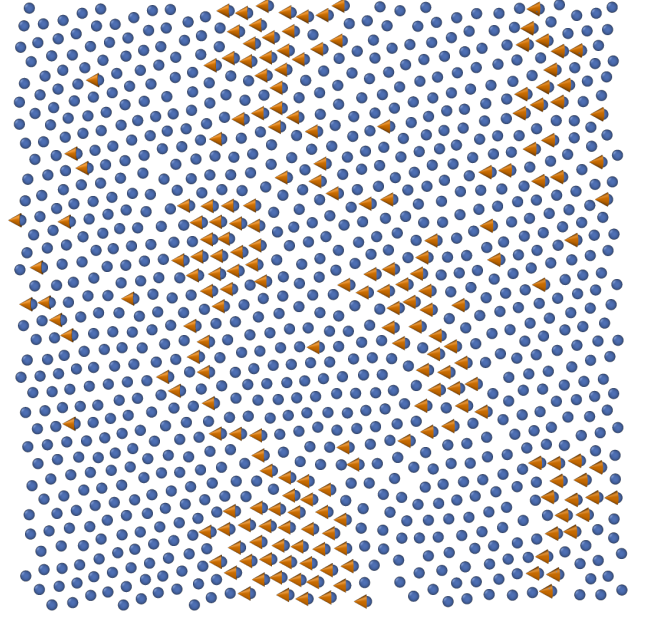


Figure 8: Visualization of clustering effects. Active particles (in orange) have a sinusoidal forcing of frequency $\omega \approx 10$ over the crossover ω_c . For visual purposes a different number of particles $N = 1024$ was chosen.

As mentioned previously, in the simulations shown in Figure 6, we considered a frequency above the crossover frequency ω_c . At frequencies below ω_c , the active particles are unable to self-organize, regardless of the forcing strength. This occurs because the passive medium has enough time to thermally relax the cages and rearrange them, allowing active particles to pass without resistance and thus removing the need for self-organization. On the other hand, for frequencies above ω_c , the timescale for the reorganization of the medium is longer than the driving force's oscillation period; therefore, the cages act as elastic solids against the active particles. As a consequence, active particles form clusters to break them. These two distinct behaviors can be observed in the visualizations in Figures 7 and 8.

3.2.2. Entropy production maximization

As established in the theoretical framework, the average dissipated power $\langle P_{diss} \rangle$ serves as a direct macroscopic proxy for the entropy production rate $\langle \dot{S} \rangle$ of the system. Figure 9 illustrates the temporal evolution of this dissipated power as the active system relaxes toward its non-equilibrium steady state (NESS).

The thermodynamic response of the system is fundamentally dictated by the density of the background medium. In the dilute, liquid-like regime ($\phi = 0.2$), the active particles do not face significant structural barriers. Consequently, their velocity remains highly correlated with the external driving force, naturally maximizing the injected mechanical work ($P \propto \mathbf{F}^{ext} \cdot \mathbf{v}$). The system reaches a highly dissipative NESS almost instantaneously, as evidenced by the flat, featureless power profile.

Conversely, in the highly dense, viscoelastic regimes

($\phi = 0.7$ and $\phi = 0.75$), the temporal evolution reveals a rich transient dynamics. Initially, the active probes are confined within the passive steric cages, severely restricting their mobility ($\mathbf{v}_i \approx 0$). This caging effect suppresses the injected power, resulting in a low initial entropy production rate. However, as the external driving persistently performs work on the medium, it triggers the structural yielding and clustering phenomena discussed in the previous section. By permanently breaking the local cages and rearranging the spatial configuration, the active particles actively increase their own mobility along the direction of the force.

This structural reorganization corresponds macroscopically to a continuous rise in the dissipated power, which eventually plateaus as the new driven NESS is attained. Crucially, the system naturally evolves from an initially constrained, low-dissipation state to a dynamically organized configuration (clustering) that allows for higher energy dissipation. As highlighted by the steepness of the transient curves in Figure 9, the rate and magnitude of this thermodynamic adaptation are strongly density-dependent. While at intermediate densities (e.g., $\phi = 0.5$) the power increment is marginal and gently reaches the plateau, highly crowded regimes ($\phi = 0.75$) exhibit a drastic and rapid jump in dissipation. The steeper slope of the initial transient at higher densities indicates a more vigorous structural reorganization: the more sterically constrained the initial state, the more aggressively the system yields and rearranges to facilitate macroscopic flow. Ultimately, this massive and rapid enhancement of the entropy production rate constitutes a striking manifestation of the MEPP [7], demonstrating that a driven viscoelastic bath dynamically restructures itself to maximize the dissipation of the injected non-conservative

work by self-organizing in clusters.

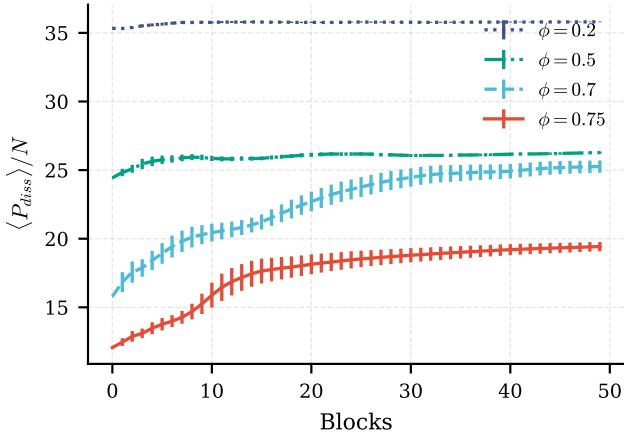


Figure 9: Average Dissipated power (per particle) as a function of the cumulative block index for different area fractions ϕ , in the active regime. Computed using block averaging (Section 2.4.1).

4. Discussion

Our simulations demonstrate a profound interplay between local structural constraints and continuous energy injection in a non-equilibrium colloidal bath. By modeling a thermal fluid with repulsive WCA forces, we mapped the medium’s transition from a simple Newtonian liquid at low densities to a strongly caged, elastic-like state at high area fractions.

Using the Mason-Weitz approximation, we identified a critical crossover frequency $\omega_c \approx 2$ at $\phi = 0.75$. At frequencies exceeding this threshold, the structural relaxation of the passive background is too slow to accommodate the forced movement of the active probes, causing the local cages to act as rigid elastic barriers. To overcome these constraints, active particles subjected to sufficiently high forcing amplitudes spontaneously self-organize into clusters. This collective behavior is notably absent at low frequencies ($\omega < \omega_c$), where thermal relaxation naturally prevents rigid barrier formation and allows single-particle drift.

Macroscopically, this structural yielding carries a distinct thermodynamic signature that strongly supports the MEPP [7]. In the highly dense regime, active particles are initially trapped, resulting in restricted mobility and a low initial rate of entropy production. However, as continuous external work forces the particles to cluster and permanently break their local cages, their mobility abruptly surges. This microscopic reorganization translates into a steep rise in macroscopic dissipated power, which eventually plateaus as the system reaches a dynamically organized NESS. Ultimately, the driven viscoelastic bath restructures itself from a sterically frustrated state into an optimal configuration that maximizes the dissipation of injected work into the environment.

5. Conclusion

In this work, we investigated the non-equilibrium dynamics of a two-dimensional WCA thermal fluid subjected to active periodic forcing. We established that when the external driving frequency exceeds the structural relaxation timescale of the dense passive medium, active particles spontaneously self-organize into clusters to overcome local elastic barriers. Crucially, we demonstrated that this collective yielding corresponds to a rapid and massive enhancement of the average dissipated power. Our findings provide compelling evidence that a driven viscoelastic bath dynamically restructures itself to satisfy the MEPP, maximizing energy dissipation into the surrounding medium.

While our overdamped Langevin model successfully captures the steric origins of this phenomenon, it inherently neglects hydrodynamic interactions, which are known to significantly influence collective motion in real dense suspensions [4]. Future investigations should explore whether this thermodynamic maximization holds when solvent-mediated forces are introduced, or when periodic forcing is replaced by persistent active propulsion (e.g., Active Brownian Particles [15]). Establishing the universality of the MEPP across different driving mechanisms and physical constraints remains an exciting avenue for advancing our understanding of active soft matter.

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